

KSG Revision-4 M1a Composite-v5 Successor Boundary

Why a published failed attempt cannot issue a receipt

pid-rs project analysis

18 August 2026

Abstract

Published commit `da253576a5f76e99633fff4de5cf1118f967b90d` is the exact composite-v4 C4 subject. Its local closure passed, but its first hosted qualification did not: CodeQL passed, repository CI contained three terminal job failures, and the dedicated v4 route failed at the static-validation preflight before substantive contract validation completed. This report records why R4 is permanently unissued, defines an append-only C5 successor with four bounded repairs and a fresh Lean replay, and permits R5 only after fresh attempt-1 success of every required route. It changes no theorem, estimator, PID object, or numerical result.

Disposition. C4 remains an immutable published subject. Its failure is retained as process evidence, not converted into partial qualification. R4 is unissued. C5 and any later R5 are new commits with new evidence. This report is not a KSG proof, PID validation, authentication, attestation, independence finding, peer review, or application gate.

1 Decision from first principles

For exact C4, define

$$Q_4 = CI_4 \wedge \text{CodeQL}_4 \wedge D_4, \quad (1)$$

where each term means terminal success at run attempt 1 for the same exact C4 commit, and D_4 is the dedicated composite-v4 route. The observations are

$$CI_4 = \text{false}, \quad \text{CodeQL}_4 = \text{true}, \quad D_4 = \text{false}. \quad (2)$$

Substitution of Equation (2) into Equation (1) gives $Q_4 = \text{false}$. The frozen issuance rule is

$$\text{issue}(R4) \iff Q_4. \quad (3)$$

Therefore R4 is permanently unissued. A passing CodeQL term cannot compensate for either failed term. A rerun would use `run_attempt = 2`; rewriting C4 would replace the published subject. Neither operation can satisfy the attempt-1 rule.

2 Exact hosted observations

Route	Run / result	Bounded interpretation
Repository CI	32079866560; failed	Three terminal job failures; the all-jobs-success term is false.
CodeQL	32079865482; passed	Actions, JavaScript/TypeScript, Python, and Rust passed; only the CodeQL term is true.
Dedicated v4	32079866461; failed	Checkout residue was rejected at the static-validation preflight; substantive validation did not complete.
Dependency Graph	32079867694; passed	Context only; neither a term in Q_4 nor a captured qualification role.

Table 1: C4 hosted observations. Route labels do not make the provider authoritative or transfer credit among routes.

The predecessor capture is retained in C5 as no-credit failure evidence. Provider responses and artifact archives are bounded observations. They do not authenticate GitHub, prove a complete provider history, or establish scientific validity.

3 Four bounded repairs

Surface	Epistemic status	C5 repair and retained boundary
Checkout residue	Exact causal reproduction	Accept absence or the one reviewed 83-byte regular, single-link image; verify SHA-256; unlink the literal entry; require absence. Every other object or byte image rejects.
Release fixture	Root cause not uniquely established by the log	Stop moving the live fixture's <code>.git</code> ; use distinct archive fixtures, phase diagnostics, and a hostile ancestor repository. Require fresh hosted requalification.
Zeta firewall	Exact reviewed byte drift	Preserve <code>uncertified</code> ; rebind the 37,869-byte section at SHA-256 <code>d1a1775d...d4d368</code> ; retain the unrelated-section negative control.
Certified SxPID	Exact stale current bindings	Rebind the final release dependency, execution-container, documentation, and support-gate bytes; retain a membership-preserving exact-line mutation.

Table 2: A local repair is necessary, not sufficient. The release-fixture row deliberately does not claim a causal diagnosis that the retained hosted log cannot establish.

The checkout normalizer never implements a general `rm`. The accepted residue SHA-256 is `443a5f645c23c3d0c0aa09f634b2ad111d46ef61946b598a2fb311678ab47454`. Symlinks, directories, hard links, altered bytes, altered permissions, and route changes fail closed.

The zeta rebind changes spelling custody only. The certified-SxPID rebind does not turn a finite exact certifier into a population theorem, a Rust refinement proof, or a general validation of shared-exclusions PID.

4 Append-only C5 and R5 topology

C5 is the exact unsigned direct child of C4, with message **Repair KSG M1a composite v5 contract**. R5, if permitted, is the exact unsigned direct child of C5, with message **Record KSG M1a composite v5 receipt**. No v4 capture or receipt path is reused.

Define

$$Q_5 = L_5 \wedge CI_5 \wedge \text{CodeQL}_5 \wedge D_5, \quad (4)$$

where L_5 is local static, hostile, replay, source-state, and publication closure; CI_5 is repository-CI all-jobs success; CodeQL_5 is success of every required CodeQL role; and D_5 is success of the dedicated composite-v5 workflow. Every term must be terminal at attempt 1 for the same exact C5 commit. The new issuance rule is

$$\text{issue}(R5) \iff Q_5. \quad (5)$$

Any false, missing, nonterminal, wrong-commit, or wrong-attempt term leaves R5 unissued and requires another append-only contract version.

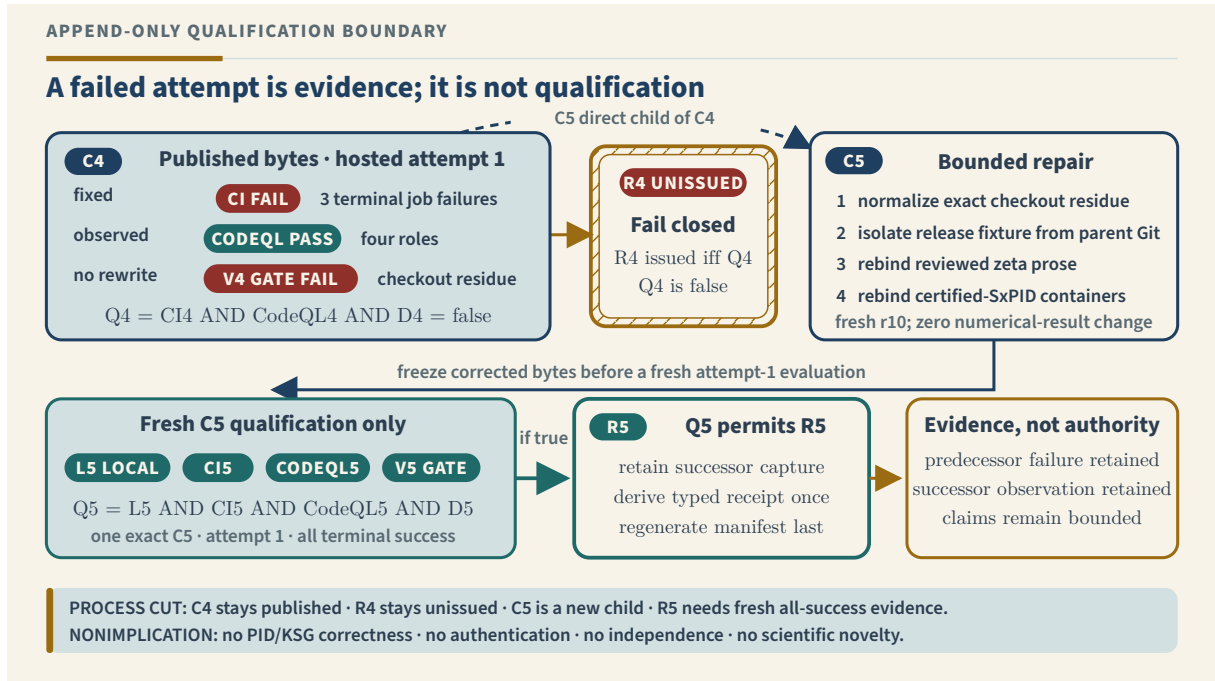


Figure 1: Append-only qualification boundary. The diagram distinguishes a retained failed attempt, a new repair commit, and a new all-success receipt route. It does not encode scientific transfer or authority.

When Q_5 is true, R5 changes exactly three paths: it adds the raw successor capture, adds the typed receipt derived from that capture, and regenerates the self-excluding current-source manifest last. The receipt hashes both the predecessor-failure capture retained in C5 and the successor-qualification capture added in R5. The manifest inventories both R5 additions while excluding itself. R5's tree and commit are outputs, never checksum inputs.

5 Replay, durability, and publication scope

C5 publishes a fresh Lean 4.33 replay as r10; accepted C4 r9 remains exact-hash-bound prior execution evidence. The suffix is numbering metadata: the tenth receipt in the 12-August

sequence and the eleventh current-project replay overall because the separate 11-August historical receipt is outside that sequence. It is not a theorem revision, proof-strength rank, or independence claim.

Remote **main** is a minimum operational recovery anchor only under provider-retention and history-rewrite assumptions. The recovery record must identify the remote URL, ref, commit OID, tree OID, parent OID, and push observation. Remote **main** is mutable, not a scholarly archive. A publication freeze should also retain the exact commit and its load-bearing predecessor and successor captures, typed receipt, replay receipt, path policy, source manifest, report and visual-review sources, figure source, and rendered PDF in a versioned release or recognized content-addressed archive, with access, licence, retention, and retrieval checks. A digest without a retrievable preimage receives commitment or bounded-omission credit only.

No branch, worktree, or temporary directory is deleted until exact-byte or semantic supersession is proved and every unique permitted payload has a remote or otherwise independent durable home.

6 Review lenses and nonclaims

The boundary is reviewed separately for logical implication, Git topology, hosted semantics, schema closure, adversarial causality, Python isolation, shell failure handling, Lean custody, PDF objects, raster layout, SVG accessibility, numerical scope, method provenance, PID-family separation, security, privacy, durability, reproducibility, usability, and publication English. Disagreement remains visible until adjudicated; a digest alone does not preserve dissent.

A separate correlated agent red-team found and corrected the original topology, run-status language, permission claim, incomplete Q_5 definitions, dual-capture binding, and durability omissions. This records review provenance but grants no dependency-disjoint, independent, or human-review credit.

This process can prevent known classes of evidence error. It does not demonstrate a causal improvement in research quality, replace peer review, or make the process itself publishable. Methods-report eligibility is necessary, not sufficient, and requires disclosure of attempt population, adaptive exposures, stopping and selection, deviations, failures, inconclusive results, resources and AI roles, dissent, human adjudication, materials, access, and licences.

Composite-v5 adds zero PID theories, zero PID functionals, zero estimators, zero theorem-source changes, and zero numerical-result changes. Nothing here transfers evidence among categorical MGW, Schick–Poland, Ehrlich continuous shared exclusions, KSG mutual information, Williams–Beer $I_{\min}$, PID2, PID3, quantized, or mixed-support routes. Nothing here establishes authentication, attestation, authorship, independence, peer review, provider completeness, application fitness, control authority, or evidence for a drone or other ecosystem deployment.